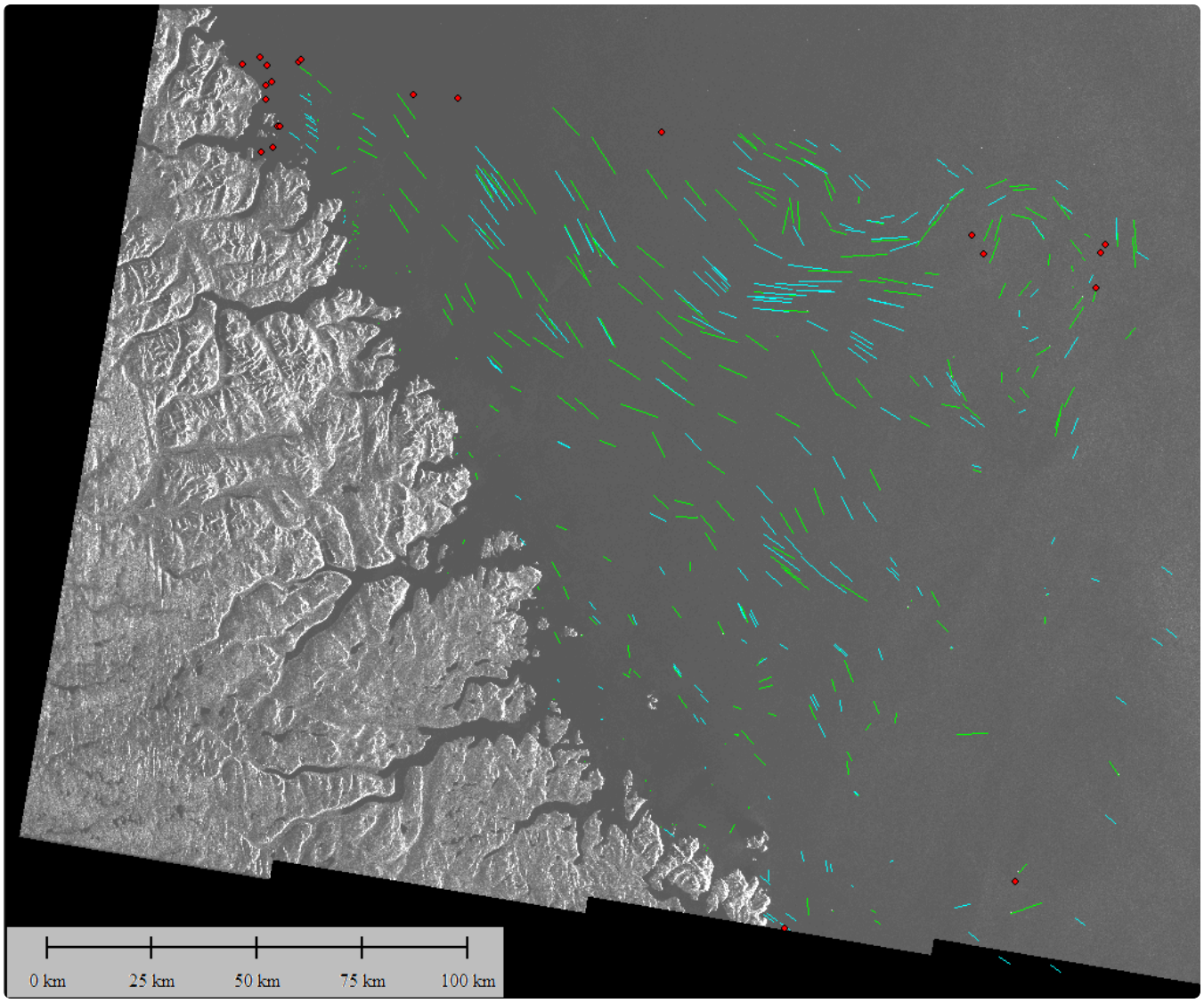
Iceberg CFAR benchmark: summary

Oct 30, 2026 · @Natali



The Sentinel-1 scene (2019-07-04) with the Færch validation labels as drift lines from the Sentinel-2 to the Sentinel-1 position. Green: verified (manually matched) icebergs; cyan: interpolated; red: unknown SAR objects.

Summary

Two different data setups. Færch et al. work on a Sentinel-1 **EW** scene at **40 m** pixel spacing (about 90 m true resolution). Our earlier THOR vs CA-CFAR work used Sentinel-1 **IW** frames at **10 m**. All results on the Færch scene below are at 40 m EW, a mode and resolution our detectors were not built on.

The published CFAR benchmark of Færch et al. (2023) is reproduced: five of its six detectors land within 0.05 of the paper's best F score, using the authors' own code and labels. This only works without thermal noise removal. The paper lists that step, but with SNAP 14 it breaks the detectors' speckle model, and every detector then stays at F 0.53 or below.

My own CA-CFAR (the baseline in the THOR vs CFAR comparison) was then run under the same protocol. It reaches F 0.587, against 0.636 for the closest published detector (gamma). It is the most precise detector on the scene (precision 0.85–0.97) but finds fewer bergs (recall ⩽ 0.45). Two of my own rules explain most of the gap:

- **The ice-strip length rule** drops any target whose grown structure is longer than 300 m. In open water it only removes real medium and large bergs. It is now applied only in scenes with pack ice.
- **The fixed 6 dB HH co-pol check** is a second threshold that doesn't move with the PFA, so it caps recall. Removing it raises F to 0.609. An HH CFAR test at the same PFA is the planned replacement.

On four of my own IIP-labelled frames, with the same windows, my CA-CFAR matches gamma: pooled recall 0.82 vs 0.83 at the same number of detections. There, too, the length rule only removes bergs, including in the two frames with pack ice.

THOR, run unchanged on the Færch scene, ties my CA-CFAR there (F 0.568 vs 0.567) at the same precision (0.83), below Wishart (0.653) and gamma (0.636). It misses faint medium bergs: those it finds stand 11 dB above the surrounding sea, those it misses 7.5 dB.

Everything here is one scene, one date and one label set. Differences under ~0.02 F are ties.

Why a benchmark

The project's headline compares THOR, a SAR foundation model with a small trained head, against my CA-CFAR. On four held-out April 2019 passes, THOR finds 0.85 of the IIP labels against 0.80 for the CFAR at the same number of detections. That comparison only means something if the CFAR is a fair baseline. It is also scored against IIP analyst labels, which are themselves drawn from SAR, so there is no independent truth.

Færch et al. (2023), *Iceberg detection in Sentinel-1 and ALOS-2 SAR imagery using CFAR detectors* (The Cryosphere 17, 5335), fills both gaps:

- six published CFAR detectors (log-normal, gamma, K, NIS, iDPolRAD, Wishart) compared on one Sentinel-1 EW scene over the Labrador Sea, 2019-07-04;
- 492 icebergs from a same-day Sentinel-2 optical image as independent truth, with drift corrected between the two acquisitions;
- open code (cfar-object-detection, GPL-3.0).

Note the difference in data. The Færch benchmark uses Sentinel-1 EW GRDM at 40 m pixel spacing (about 93 × 87 m resolution). Our earlier work used Sentinel-1 IW frames at 10 m (about 20 m resolution), labelled by IIP. The two sets of numbers are therefore not compared directly (see Caveats).

The plan was to reproduce the paper first, then run my CA-CFAR under exactly the same protocol. Only the C-band part is in scope (the ALOS-2 L-band scene is not). The GPL library runs in its own environment and talks to my code through files only, so none of it is copied into this repo.

Data

Everything except the validation labels is public. The labels came from Laust Færch by email on 2026-10-24 and cannot be rebuilt from the paper alone.

Item	What was used	Where from	Access
SAR scene	Sentinel-1A EW GRDM, HH/HV, 2019-07-04 10:27:30 (`…_027966_032866_4053`)	Copernicus Data Space (CDSE)	Public, free login
SAR, comparison only	Same pass as EW GRDH (`…_DEF7`)	ASF	Public, free Earthdata login
Optical scene	Sentinel-2B L1C, 2019-07-04 15:36, 8 tiles, original baseline N0207	Google public Sentinel-2 bucket	Public. CDSE only has the N0500 reprocessing
Land mask	OSM coastline ways, Newfoundland & Labrador extract (.osm.pbf), buffered 500 m	OpenStreetMap	Public. osmdata.openstreetmap.de was unreachable

Item	What was used	Where from	Access
Detector code	cfar-object-detection, commit `4136b4c`	GitHub / Zenodo	Public, GPL-3.0
Paper text and tables	Full-text JATS XML	The Cryosphere	Public, CC BY 4.0
Validation labels	492 drift lines (S2 position → S1 position), 284 `manual` / 208 `interpolate`, plus 22 "unknown" SAR objects	Laust Færch, by email	Private. Kept in `data/`, never committed

What the paper leaves open

The paper is thorough, and its open code and labels are what made this reproduction possible. A few details still had to be worked out from the data, the code or the labels. None of them changes the paper's conclusions; they are listed here in case they help anyone reproducing the study from the text.

Details where my data looked different

#	In the paper	What I found in the data	What I did
1	Table 1: product is EW **GRDH**	The 40 m spacing and ENL 10.7 match the GRDM product definition. On this scene GRDH gives 575 false positives against 341 for GRDM (gamma, PFA 1e-19)	Used GRDM, assuming "GRDH" in Table 1 may be a small slip
2	Table 1: S1 orbit 028068	The product at the paper's acquisition time seems to have absolute orbit 027966 (relative 069)	Identified the product by time and footprint
3	Table 1: S2 relative orbit R068	The tiles I found are relative orbit R111	As above

#	In the paper	What I found in the data	What I did
4	Table 1: five S2 tiles	These cover about 52 % of the iceberg area; Fig. 1′s S2 outline seems to need three more (T20VNK, T20VMK, T20VML) to reach 99 %	Used eight tiles
5	Thermal noise removal is part of the SNAP chain	With SNAP 14 this step changes the ENL the detectors rely on, and only the run without it reproduces the paper′s results (next section). An older SNAP version may well behave differently	Ran the reproduction without it

Details I had to assume

#	Detail	What I found or assumed
6	SNAP version	Not given. SNAP 14 also ignores `removeThermalNoise=false`, so I removed the node from the graph
7	Units of berg length and drift	They appear to be EPSG:5938 polar-stereographic metres (1.07× true metres at 58° N); with this factor the size classes match Table 2 exactly (181 / 175 / 136). In true metres: 196 / 175 / 121
8	Search radius for the 15 small `manual` labels	I used 250 m. The 250 / 1789 m radii may also be in EPSG:5938 metres (~234 / 1671 m true)
9	ENL passed to gamma and Wishart	Table 1 gives one value (10.7) for both bands. Gamma and Wishart reach the paper′s F 13–15 decades of PFA higher, so the effective ENL in the original runs may have been different
10	Study area	Shown in Fig. 1; I georeferenced it from the figure (see Data)

#	Detail	What I found or assumed
11	Window sizes	iDPolRAD's 57 px training window is above the library's 41 px limit, so I assume it was cropped. The window for the Sentinel-2 4-sigma detector isn't given; I used the SAR window

Points 5, 6, 8 and 9, and the different optimum PFAs in Tables 3 and 5, are the ones I'd most like to check with Laust Færch.

Processing, and the thermal noise finding

Run as the paper describes, with thermal noise removal, no detector came within 0.05 of the paper. Without it, five of six did. The cause is how SNAP 14 removes the noise floor.

Processing chain (SNAP 14 graph, from the paper's footnote). Apply orbit file → GRD border-noise removal → thermal noise removal → calibration to sigma0 (HH, HV) → ellipsoid correction to 40 m. Details the paper doesn't give: Ellipsoid-Correction-RD (the GG variant has no pixel-spacing setting), nearest-neighbour resampling, UTM 20N, and `nodataValueAtSea=false`, without which the whole sea is set to no-data. The output stays in linear intensity; dB is only for display.

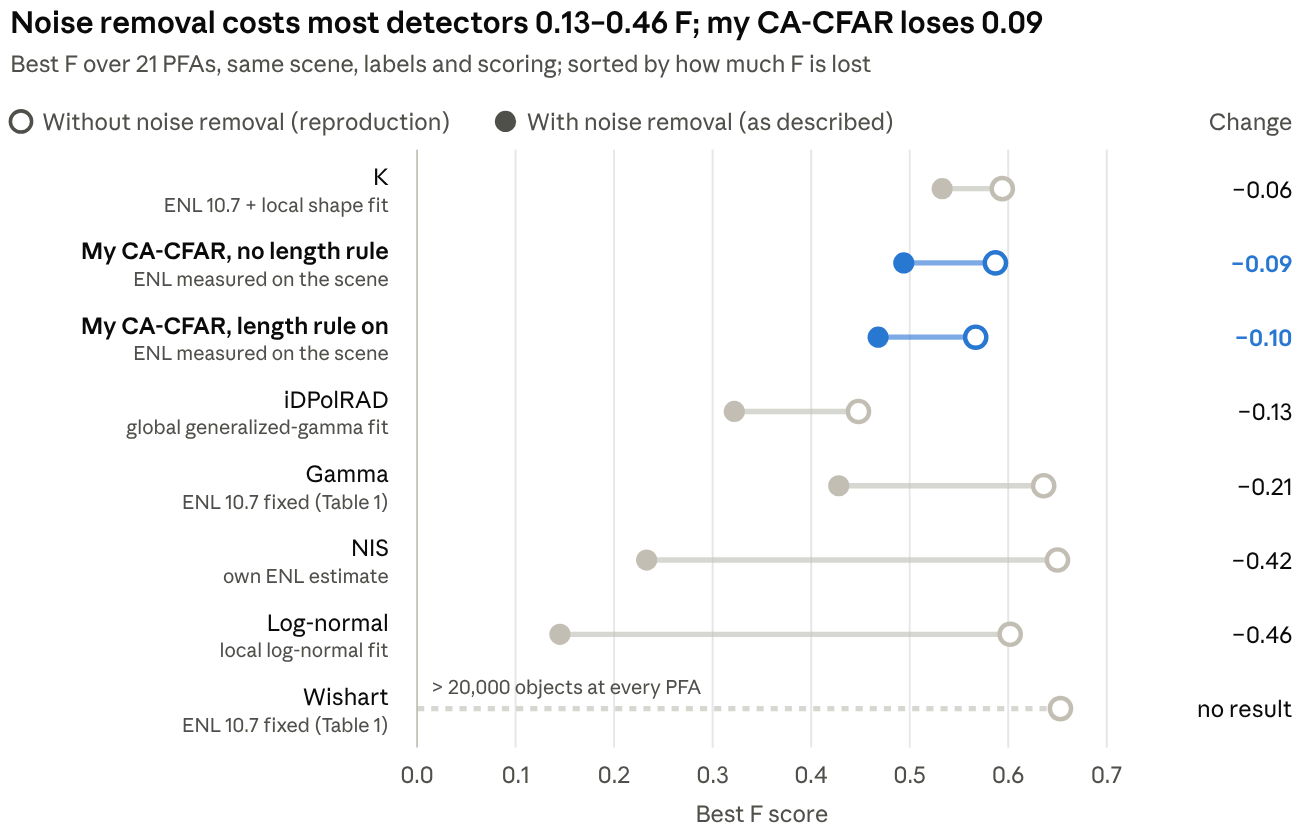
Two terms. *ENL* (equivalent number of looks) measures how smooth the speckle is. A CFAR sets its threshold from it, so a wrong ENL means a wrong false-alarm rate. *Thermal noise removal* subtracts the sensor's noise floor, which matters most in the weak cross-pol (HV) channel.

What goes wrong. SNAP 14 subtracts the noise floor and clips negative results to ~3.7e-12 (−115 dB) instead of 0. In open water, 23.7 % of HV pixels and 0.75 % of HH pixels end up at that value. Dark, calm water then fluctuates far more than an ENL of 10.7 allows:

ENL on the 40 m GRDM stack	HH	HV
Table 1 (product definition, used by the paper)	10.7	10.7
Without thermal noise removal, median 2 km open-water box	10.74	11.45
With thermal noise removal, same boxes	9.5	0.98
With, 1 km boxes around gamma's false positives	0.51	0.43

Without noise removal the measured ENL matches Table 1, which is what the detectors assume. With it, false positives cluster in one dark patch near the coast (median HH −35.6 dB).

Decision (2026-10-26). The run without thermal noise removal is the reproduction run. The run as described stays in the report as a grey curve. Which SNAP version the paper used, and what its noise removal did, is an open question for the authors.

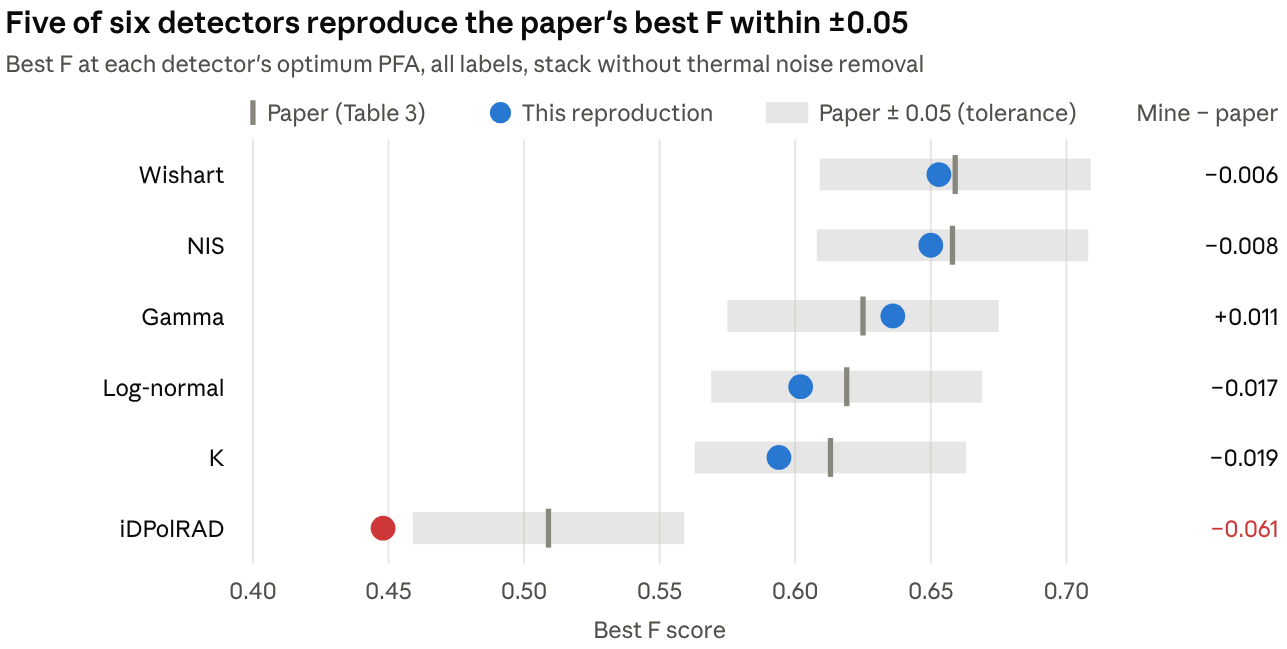


bench-score: scores_noTNR.csv and scores_TNR.csv, Færch labels (Path A), all labels, scoring area fig1

The ranking flips with noise removal. Detectors that take the ENL from the product definition (gamma, Wishart) or whose model breaks on the clipped pixels lose the most. K and my CA-CFAR adapt to the scene's statistics and keep most of their F. NIS estimates its own ENL yet still collapses (1,085 false positives); not investigated. Gamma, NIS and iDPolRAD peak at 1e-21, the end of the sweep, so their F with noise removal may be slightly underestimated.

Reproduction results

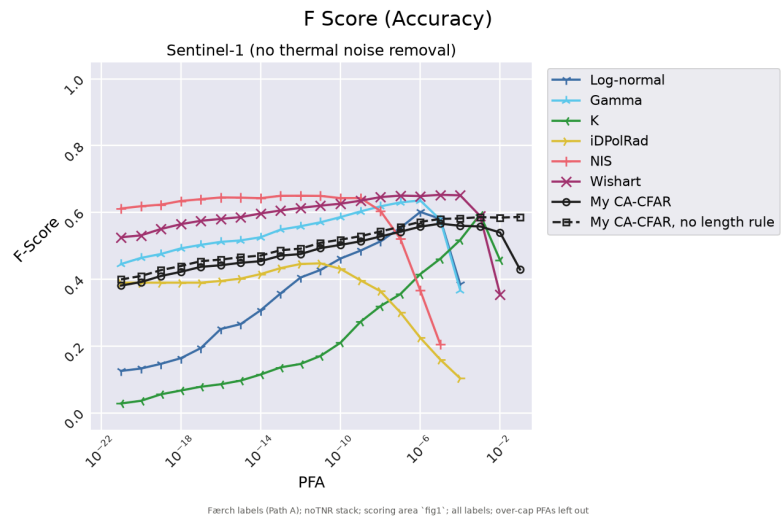
Five of six detectors reproduce the paper's best F within 0.05: gamma +0.011, Wishart −0.006, NIS −0.008, log-normal −0.017, K −0.019. iDPolRAD is 0.061 below. Setup: the paper's code, windows of 360 / 520 m (9 / 13 px), HH AND HV merging, ENL 10.7, 21 PFA levels from 1e-21 to 1e-1, labels from Færch, no thermal noise removal.



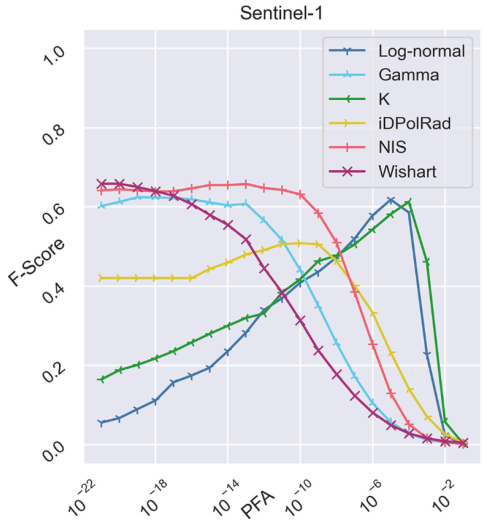
Paper: F derived from Table 3 (TP, FP, FN). Mine: bench-score, scores_noTNR.csv, Færch labels (Path A)

The curve shapes agree with the paper. Its sec. 4: "The K detector and the log-normal detector achieve their maximum accuracy at a very narrow interval of PFA levels, whereas the gamma, iDPolRAD, NIS, and Wishart generally exhibit higher F score across a wide range of PFA levels."

(a) This reproduction



(b) Færch et al. (2023), Fig. 6



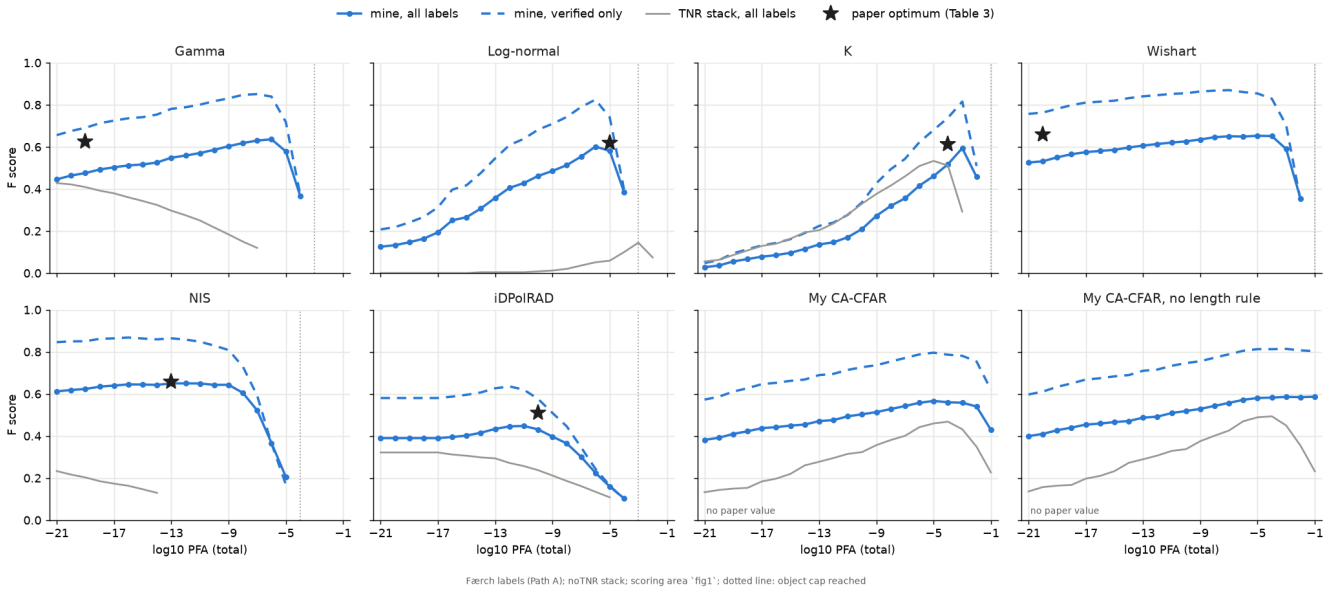
F score vs PFA, all labels. (a) This reproduction, stack without thermal noise removal; my CA-CFAR in black. (b) Færch et al. (2023), Fig. 6, CC BY 4.0.

Where it differs.

- *Optimum PFA.* NIS, K, log-normal and iDPolRAD peak within one decade of the paper. Gamma and Wishart reach the paper's F, TP and FP 13–15 decades higher. They are the two detectors that take the ENL as given, so a different effective ENL in the paper's runs would shift the PFA axis without changing the F level. Not checked.
- *iDPolRAD.* Same false positives as the paper (70 vs 74), fewer true positives (162 vs 193), mostly medium bergs. Its threshold comes from one generalized-gamma fit over the whole image, so it depends on what the image contains: land mask, swath edges and noise-floor handling all differ from the paper's.

What holds.

- The labels reproduce Table 2 exactly (181 / 175 / 136) and the mean verified drift (3,954 vs 3,953 m).
- Detected share by size follows Table 5: large 87–93 %, small 4–10 % (iDPolRAD excepted).
- Verified-only F is 0.81–0.87 at each optimum. Most misses are interpolated labels, i.e. small bergs whose position is uncertain to ~1.8 km.
- 117 of 126 (detector, PFA) runs were scored. The other 9 produced more than 20,000 objects, where precision is below 0.1.



Per detector: solid blue = all labels, dashed = verified labels only, grey = the stack with thermal noise removal, star = the paper's optimum (Table 3), dotted line = object cap reached.

What we learned about my CA-CFAR

My CA-CFAR is precise but conservative: on the Færch scene it reaches F 0.587 against gamma's 0.636, with precision 0.85–0.97 over the whole sweep and recall never above 0.45. Two post-processing rules from Stage 1, not the CFAR core, explain most of that gap.

How it differs from the published detectors. It runs a cell-averaging CFAR on HV only, with the threshold from the measured scene ENL (13.4 here). Then four rules act on its output: an HH co-pol check (6 dB), merging of targets by grown structure, a 200 m buffer from the mask edge, and the ice-strip length rule (300 m). The published detectors instead run the CFAR test on HH and on HV and keep a pixel only if both pass (HH AND HV, per-channel PFA = √total PFA). They take the ENL from the product definition and have no shape rules.

On the Færch scene	Best F (PFA)	TP / FP / FN	Precision	Small / medium / large found
Gamma (paper's code)	0.636 (1e-6)	255 / 55 / 237	0.82	9.9 / 64.6 / 91.2 %
My CA-CFAR, length rule on	0.567 (1e-5)	200 / 14 / 292	0.93	3.3 / 47.4 / 81.6 %

On the Færch scene	Best F (PFA)	TP / FP / FN	Precision	Small / medium / large found
My CA-CFAR, length rule off	0.587 (1e-1)	220 / 38 / 272	0.85	6.1 / 52.0 / 86.8 %

1. The ice-strip length rule removed real bergs

The rule drops any target whose grown structure (pixels 4 dB over the background around it) is longer than 300 m. It was tuned on a 2025 scene with pack ice, where floe strips form long bright structures. In open water it only removes icebergs, mostly the medium and large ones:

- **Færch scene (no pack ice):** −0.02 F, large bergs found 82 % vs 87 %. At PFA 1e-1 it costs 0.16 F, because the low threshold chains speckle into long structures that the rule then drops.
- **IIP frame 2019-04-29 at 10 m:** recall at 500 detections 0.75 with the rule vs 0.83 without. The rule cost 21 medium bergs and all 4 large ones.
- **THOR comparison (Stage 1):** the CFAR found only 4 of 14 large bergs. Their grown blobs were 340–470 m long.

Fix (2026-10-26). The rule now applies only when a scene's pack-ice mask covers at least 500 km² (`detections.length_rule: auto`). Our scenes are either ⩽ 300 or ⩾ 1,200 km². The THOR vs CFAR headline moved from 0.84 vs 0.74 to 0.85 vs 0.80, and large bergs from 11 vs 4 to 11 vs 10. The 500 km² cutoff was set after seeing 04-29 (no labels used), and that caveat goes with the headline. The four-frame run (next section) shows the rule costs recall in the icy frames too, in the area that is scored. Dropping it everywhere is now an open decision. Working with sea ice in the scenes needs more study and research before a better rule can replace it.

2. Measuring the ENL is its strength

On the stack with thermal noise removal, my CA-CFAR measures the HV ENL as 1.05 (13.4 without), and its threshold rises to match. It loses 0.09 F, against 0.21 for gamma and Wishart's total collapse, which keep ENL 10.7. What it still loses comes from the clipped −115 dB pixels, which pull the local background mean towards zero: false positives rise from 14 to 81.

How to improve it

1. **Replace the 6 dB check with an HH CFAR test** at the same per-channel PFA (HH AND HV, like gamma), keeping the measured ENL per band. Expected: close to gamma on clean data, and the robustness with noise removal kept. To test next.
2. **Study and research ice in the scenes:** how the literature separates icebergs from pack ice and other sea ice, before deciding what should replace the length rule.

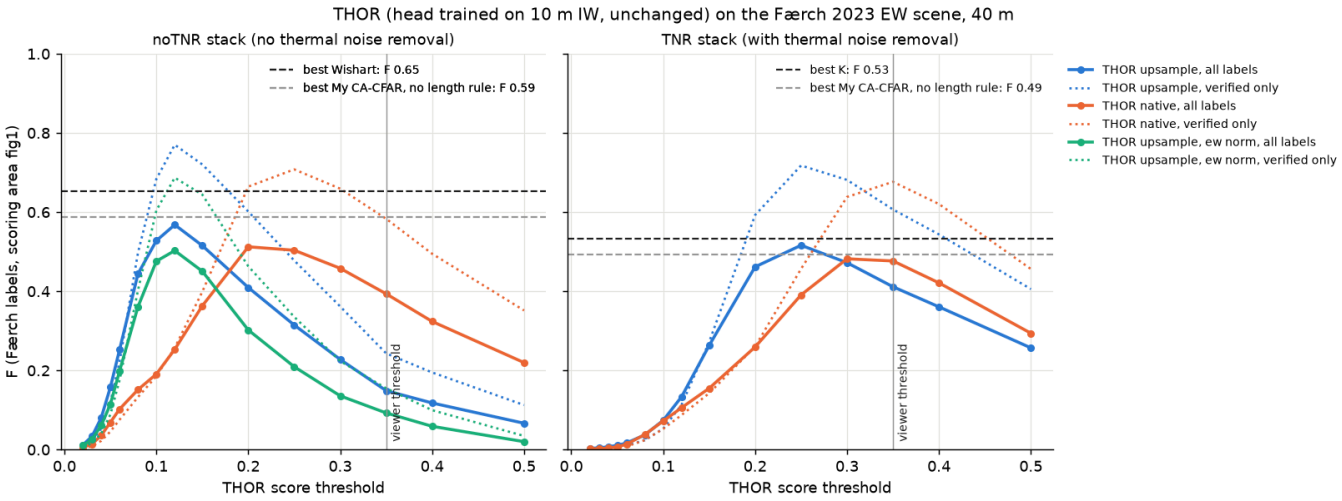
THOR on the Færch scene

The project's THOR head was also run on the Færch scene, unchanged. It was trained on 10 m HyP3 IW frames off Newfoundland in April–June and never saw EW mode, 40 m pixels or July. Nothing was retrained; only the score threshold is swept, the way the CFARs' PFA is. Scoring is the same as for the CFARs: Færch's labels, the same area and the same radii. The IIP frames have no precision, so this is the first precision number for THOR.

Detector (no thermal noise removal, best F)	Detections	Precision	Recall	F	Recall small / medium / large
Wishart	329	0.82	0.55	0.653	0.09 / 0.72 / 0.93
Gamma	310	0.82	0.52	0.636	0.10 / 0.65 / 0.91
My CA-CFAR, no length rule	258	0.85	0.45	0.587	0.06 / 0.52 / 0.87
THOR	258	0.83	0.43	0.568	0.06 / 0.49 / 0.86
My CA-CFAR	214	0.94	0.41	0.567	0.03 / 0.47 / 0.82

- **THOR ties my CA-CFAR here** and stays below the best published detectors. The IW headline (recall 0.85 vs 0.80 at matched counts) doesn't carry over to an unseen mode, resolution and month.
- **Where it loses: medium bergs.** Recall 0.49, against 0.65 for gamma and 0.72 for Wishart. Its precision matches the CFARs'. Small bergs are lost by every detector at 40 m.
- **Thermal noise removal costs THOR 0.05 F** (0.516), against 0.21 for gamma: it has no ENL to break. With noise removal it is second, behind K (0.533).
- **Input handling.** Resizing the 40 m crop to the 10 m grid THOR was trained on beats 2 × 2 px patches at 40 m (0.568 vs 0.512), mostly through precision. THOR's own EW normalisation is worse than the IW one the head was trained with (0.503).

- **Scores don't transfer between products.** The best threshold is 0.12 here (0.25 with noise removal), against 0.35 in the viewer.



F vs THOR score threshold. Left: no thermal noise removal; right: with it. Dashed lines: the best published detector and my CA-CFAR on the same stack; dotted: verified labels only.

Why THOR misses medium bergs

The misses are faint targets. A medium berg THOR finds stands a median 11.0 dB above the surrounding sea in HH; one it misses, 7.5 dB. No THOR setting recovers them.

Medium bergs, HH contrast to the surrounding sea	Bergs	THOR recall
< 4 dB	4	0.00
4–8 dB	41	0.49
> 8 dB	83	0.73

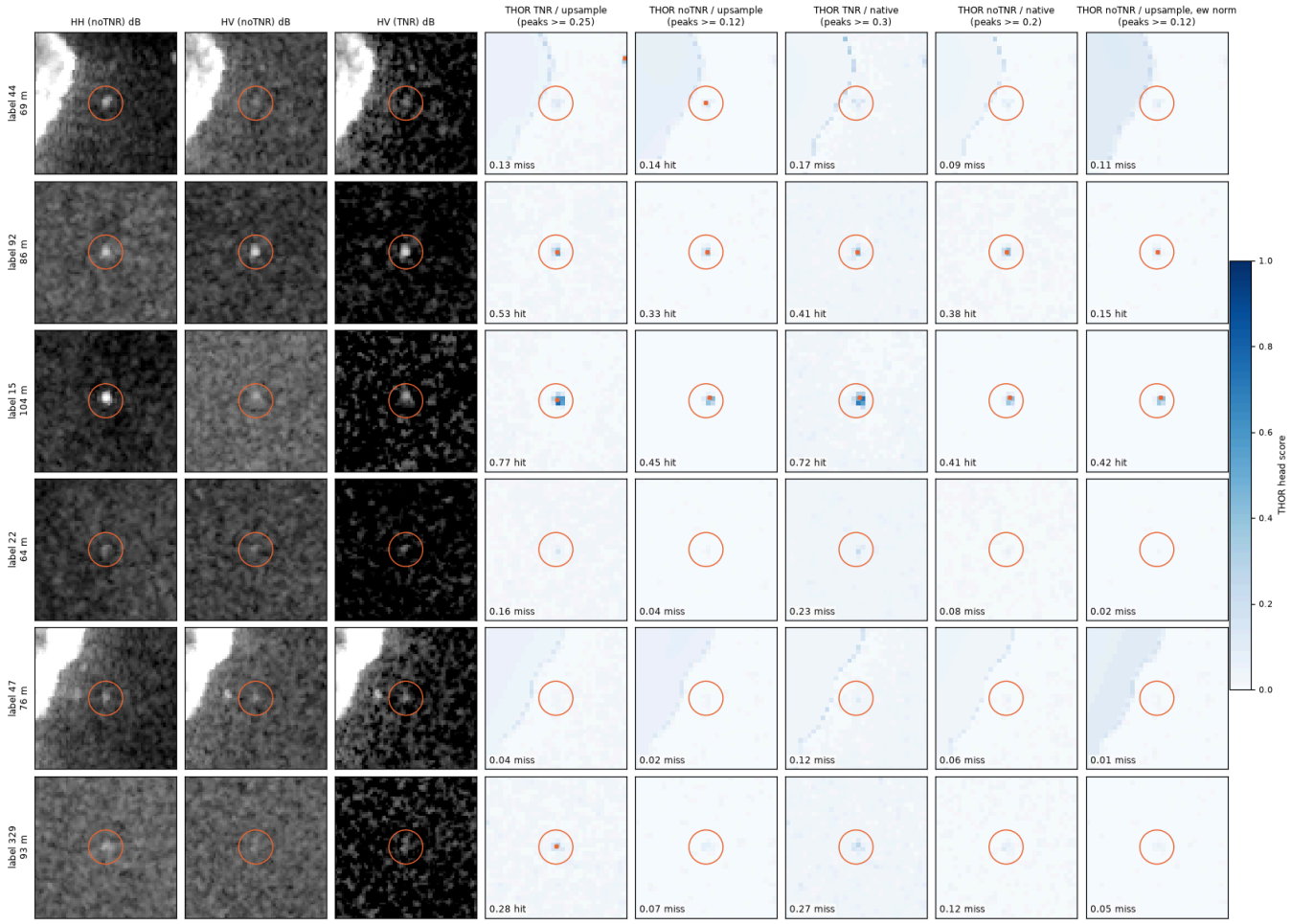
Verified medium labels in the scoring area, THOR without noise removal at its best threshold (0.12); a hit is a THOR peak within 250 m, a looser count than the table above. Incidence angle doesn't explain the misses: recall is 0.61 at 35–40° and 0.58 at 40–50°.

- **EW resolution is about 90 m, not 40 m.** The 40 m is pixel spacing; the Sentinel-1 product specification gives EW GRDM about 93 × 87 m, against about 20 m for our IW frames. A 60–120 m berg fills about one resolution cell, its echo mixed with sea clutter.
- **Resolution alone costs score, not detection.** Training bergs averaged to 40 m drop from 0.49–0.84 to 0.28–0.78 and stay well above the water. That 4 × 4 average blurs about half as much as the real EW product.

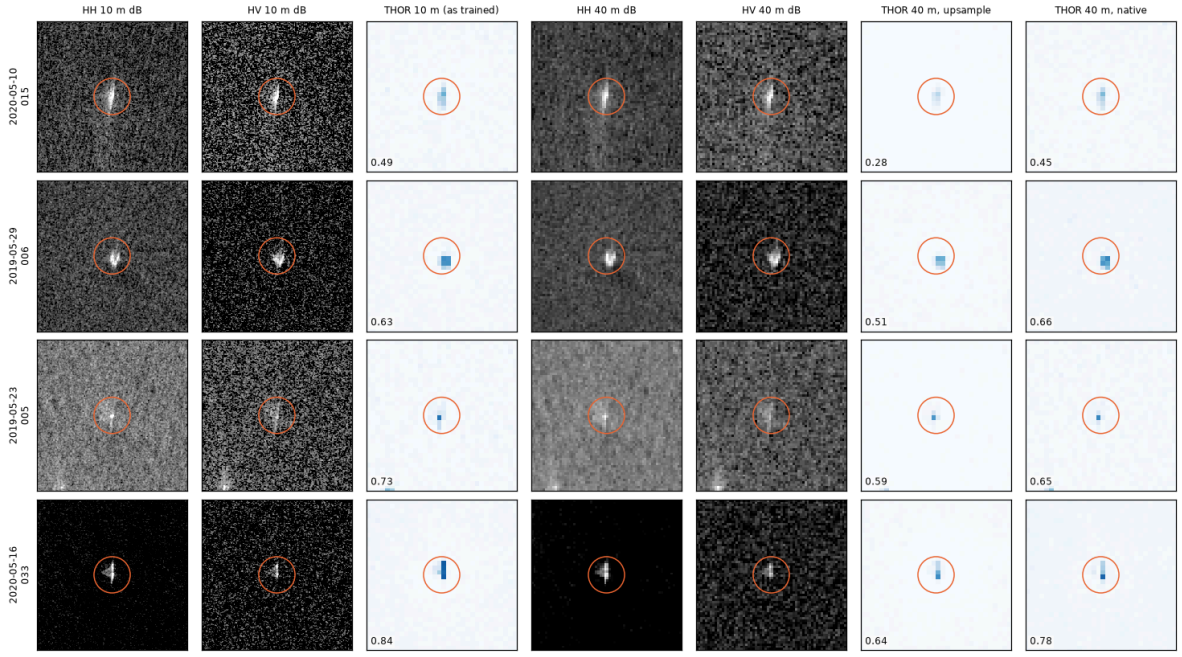
- **The labels differ.** IIP analysts labelled the training bergs in the SAR image itself, so those bergs stand out by construction. Færch's labels come from Sentinel-2 and include bergs that are faint in SAR. IIP "MED" (61–120 m) and Færch "medium" (60–120 m) share a size band, but the training bergs look far larger and brighter.

The CFARs face the same physics and still reach 0.65–0.72 recall on medium bergs. A CFAR asks whether a pixel is brighter than its local sea by a margin; THOR asks whether a patch looks like the bergs it was trained on, and a faint dot doesn't.

Medium icebergs through THOR (linear head, unchanged). Panels 2.56 km; ring = label; squares = peaks at the run's threshold; score = best token within 250 m
Færch 2023 scene (EW, 40 m, July): verified medium bergs, hits then misses of noTNR / upsample at its best-F threshold



Training chips (IW, 10 m, April-June): IIP medium bergs (61-120 m) at spread-out training scores; the same chip averaged to 40 m on the right



Top: six verified Færch medium bergs, three hits and three misses of the best run, under every THOR run; the number is THOR's score at the berg. Bottom: four training medium bergs at 10 m, then the same chip averaged to 40 m. Panels 2.56 km wide; one score scale for all heatmaps.

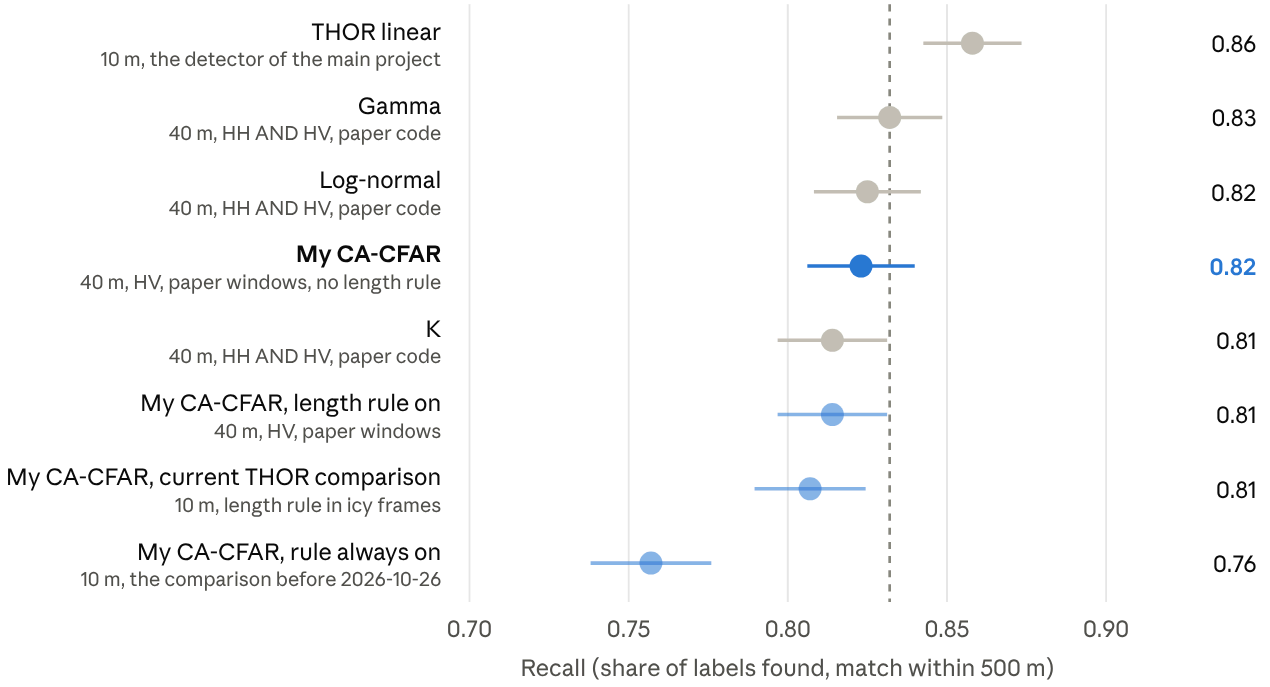
On four of my own frames, and what's next

On four of my IIP-labelled frames, my CA-CFAR finds the same share of icebergs as the published gamma CFAR: 0.82 vs 0.83 of 512 labels, pooled at the same number of detections. The frames are the four held-out April 2019 passes of the THOR comparison (04-17, 04-23, 04-24, 04-29), HyP3 RTC HH/HV, multilooked 4×4 to 40 m so the paper's windows fit the library. Every detector uses the paper's 360 / 520 m windows. Two frames have pack ice; detections within 5 km of it are not scored. Precision can't be measured without ship positions (AIS), so detectors are compared by recall at the same number of detections: 2 per label, i.e. 1,024 pooled.

On four IW frames my CA-CFAR matches gamma: 0.82 vs 0.83

Pooled recall at 1,024 detections (2 per label), 512 same-pass IIP labels, April 2019

Whiskers: ±1 standard error. Dashed line: gamma, the closest published detector.



bench-iw-compare: summary_recall.csv (pooled column), four frames 2019-04-17 / 23 / 24 / 29

- **Per frame** my CA-CFAR is within 0.03 of gamma everywhere: 04-17 0.61 vs 0.57, 04-23 0.82 vs 0.81, 04-29 0.85 vs 0.88. On 04-24 (calm open water) it never produces more than 87 detections and has found all 49 labels by then. Large bergs pooled: 13 of 14 for both.
- **Window size doesn't matter pooled.** With its own Stage 1 windows (200 / 840 m) it reaches 0.83.
- **NIS finds the most bergs where it runs** (0.88 at 1,082 detections, and the most small bergs), but even at PFA 1e-21 it can't produce fewer detections than that. **Wishart is the opposite:** it never passes 932 detections, because it takes one ENL for both bands.
- **iDPolRAD looks weak pooled (0.66) only because of how frames are pooled.** Its threshold comes from one fit over the whole image, so the same PFA floods one frame (up to 107,173 detections on 04-17) and stays strict in the others. Per frame it is level with the rest.
- **THOR (10 m) leads every CFAR pooled by 0.03–0.05.** It leads on 04-23 and 04-29, is second on 04-24 (NIS 1.00) and mid-pack on 04-17.
- **The length rule also costs recall in the icy frames.** The 5 km filter already removes the ice-edge strips the rule was meant to catch, so in the scored area it only removes bergs. At 10 m on 04-17 the CFAR with the rule stalls at 0.67, against 0.83 without. Pooled, the current THOR comparison's CFAR (rule on in icy frames) is at 0.81, and the same CFAR with the rule off everywhere at 0.82. Whether the rule helps within 5 km of the ice is not measured.

Caveats

One scene, one date and one label set for the reproduction. Four IW frames from one month for the check on my data (512 labels; two frames hold under 50 labels each). The EW GRDM scene (sigma0, 40 m) and the IW RTC frames (gamma0, 10 m multilooked to 40 m) differ in resolution, ENL and radiometry, so their numbers are not compared directly.

Next steps

1. **Ground truth and validation**
    - Establish reliable ground truth and validation data, starting from the reference datasets and validation methods of published studies
    - Focus on datasets and geographic regions where some form of independent validation is available: for example- same-day optical imagery for icebergs, as in Færch et al., and AIS records for ships

- Review the Statoil/C-CORE Iceberg Classifier Challenge dataset for ship vs iceberg discrimination, keeping in mind the data-leakage concerns some publications have noted
- Investigate the dataset described in *SAR Ship-Iceberg Discrimination in Arctic Conditions Using Deep Learning*;

2. **Data availability**
   - Analyze which types of SAR and optical (EO) satellite scenes are available over those regions, and at which frequency
3. **Literature research**
   - Ice detection: how icebergs are separated from pack ice and other sea ice
   - Ship detection, and ship vs iceberg discrimination